$$g(x)=-1-x+\ln x+t$$

$$g\left(x_1\right)=g\left(x_2\right)=0$$

证

$$x_1+x_2>t$$

证:

$$\begin{aligned}&g\left(x\right)=0\\&\Leftrightarrow -1-x+\ln x+t=0\\&\Leftrightarrow x+1-\ln x=t\end{aligned}$$

如图 1

易得 $t>2$, 不妨设 $0<x_1<1<x_2$

$$\begin{aligned}&x_1+x_2>t\\\Leftrightarrow&x_1+x_2>x_1+1-\ln x_1\\\Leftrightarrow&x_2+\ln x_1>1\end{aligned}$$

设

$$x_1=x_1\left(t\right),x_2=x_2\left(t\right)$$

$$h\left(t\right)=x_2+\ln x_1$$

则只需证

$$h\left(t\right)>1$$

$$\begin{aligned}g\left(x_1\right)&=-1-x_1+\ln x_1+t=0\\g\left(x_2\right)&=-1-x_2+\ln x_2+t=0\end{aligned}\tag{1}$$

即

$$\begin{aligned}t&=1+x_1-\ln x_1\\t&=1+x_2-\ln x_2\end{aligned}\tag{2}$$

两侧对 $t$ 求导

$$\begin{aligned}1&=\left(1-\frac{1}{x_1}\right)x_1'\left(t\right)\\1&=\left(1-\frac{1}{x_2}\right)x_2'\left(t\right)\end{aligned}\tag{3}$$

$$\begin{aligned}x_1'\left(t\right)&=\frac{1}{1-\frac{1}{x_1}}=\frac{x_1}{x_1-1}\\x_2'\left(t\right)&=\frac{1}{1-\frac{1}{x_2}}=\frac{x_2}{x_2-1}\end{aligned}\tag{4}$$

$$h\left(t\right)=x_2+\ln x_1$$

对 $t$ 求导

$$\begin{aligned}h'\left(t\right)=&x_2'\left(t\right)+\frac{x_1'\left(t\right)}{x_1}\\=&\frac{x_2}{x_2-1}+\frac{\frac{x_1}{x_1-1}}{x_1}\\=&\frac{x_2}{x_2-1}+\frac{1}{x_1-1}\\=&\frac{x_1x_2-1}{\left(x_1-1\right)\left(x_2-1\right)}\end{aligned}\tag{5}$$

已得 $x_1<1,_2>1$, 下证明 $x_1x_2<1$

$$\begin{cases}t=1+x_1-\ln x_1\\t=1+x_2-\ln x_2\end{cases}$$

上式减下式得

$$x_2-x_1=\ln x_2-\ln x_1=\ln\frac{x_2}{x_1}$$

设 $p=\frac{x_2}{x_1}$, $p>1$

$$\begin{cases}x_2-x_1=\ln p\\p=\quad\frac{x_2}{x_1}\end{cases}$$

得

$$\begin{cases}x_1=\quad\frac{p}{p-1}\ln p\\x_2=\quad\frac{1}{p-1}\ln p\end{cases}$$

$$\begin{aligned}x_1x_2=&\frac{p}{\left(p-1\right)^2}\ln^2 p\\=&\left(\frac{\sqrt{p}}{p-1}\ln p\right)^2\end{aligned}\tag{6}$$

下证 $\frac{\sqrt{p}}{p-1}\ln p<1$

$$\begin{aligned}&\frac{\sqrt{p}}{p-1}\ln p<1\\&\Leftrightarrow \ln p<\frac{p-1}{\sqrt{p}}\\&\text{设}q=\sqrt{p},\quad q>1\\&\Leftrightarrow 2\ln q<\frac{q^2-1}{q}\\&\Leftrightarrow 2\ln q-\frac{q^2-1}{q}<0\end{aligned}\tag{7}$$

设 $m(q)=2\ln q-\frac{q^2-1}{q}$

$$\begin{aligned}m'(q)&=2\frac{1}{q}-1-\frac{1}{q^2}\\&=\frac{2q-q^2-1}{q^2}\\&=-\frac{(q-1)^2}{q^2}<0\end{aligned}$$

$$\begin{aligned}&\therefore m(q)\text{在}(2,+\infty)\downarrow\\&\because q\to 2,m(q)\to 0\\&\therefore q>0,m(q)<0\\&\therefore \frac{\sqrt{p}}{p-1}\ln p<1\\&\text{由}(6),x_1x_2<1\end{aligned}$$

$$\begin{aligned}&\text{由}(5),h'(t)=\frac{x_1x_2-1}{(x_1-1)(x_2-1)}\\&\because x_1<1,x_2>1,x_1x_2<1\\&\therefore h'(t)>0\\&h(t)\text{在}(2,+\infty)\uparrow\\&\text{当}t\to 2\\&x_1\to 1,x_1\to 1,\ln x_1\to 0\\&h(t)=x_2+\ln x_1\to 1\\&\therefore t>2,h(t)>1\\&\text{即证}\end{aligned}$$

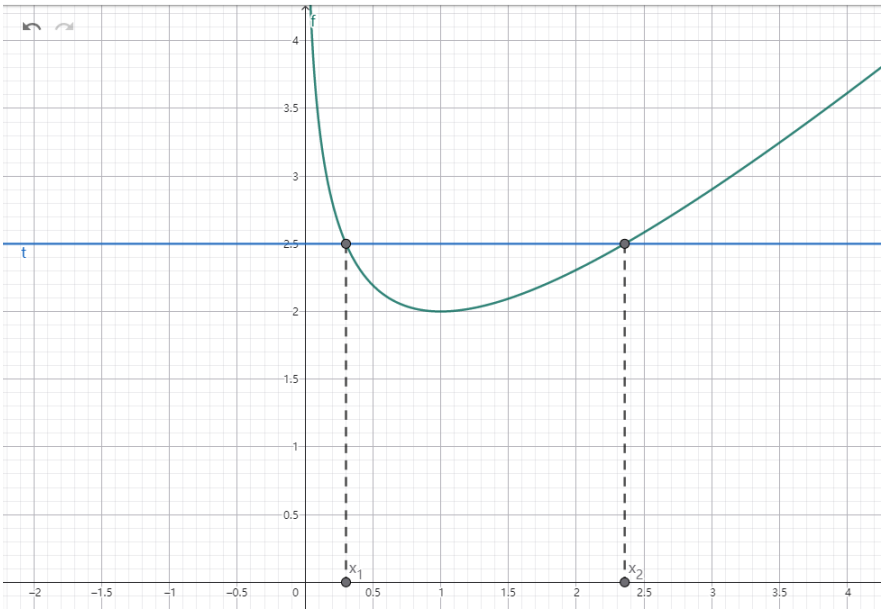


图 1: $y = x - \ln x + 1$

$y = t$